AQ7.2: Activity Questions 2 - Not Graded

This assignment will not be graded and is only for practice.

Level 1

1 point

Let LL be an affine subspace of R^3R3 defined as $L = \{(x, y, z) \mid x + y + 2z = 6\}$L = {(x, y, z) | x + y + 2z = 6}, then which of the following subspaces of R^3R3 corresponds to the affine subspace LL?

- $\{(x, y, z) \mid x + y + z = 5\}$ {(x, y, z) | x + y + z = 5}
- $\{(x, y, z) \mid x + y + 2z = 0\}$ {(x, y, z) | x + y + 2z = 0}
- $\{(x, y, z) \mid x + 2y + z = 0\}$ {(x, y, z) | x + 2y + z = 0}
- $\{(x, y, z) \mid 2x + y + z = 0\}$ {(x, y, z) | 2x + y + z = 0}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(x, y, z) \mid x + y + 2z = 0\}$ {(x, y, z) | x + y + 2z = 0}

1 point

Which of the following subsets of R^2R2 represent an affine subspace of R^2R2?

- [ ] $\{(x, y) \mid x^2 + y^2 = 0\}$ {(x, y) | x2 + y2 = 0}
- [ ] $\{(x, y) \mid x^2 + y^2 = 4\}$ {(x, y) | x2 + y2 = 4}
- [ ] $\{(x, y) \mid x^5 + 1 = y\}$ {(x, y) | x5 + 1 = y}
- [ ] $\{(x, y) \mid 3x + y = 1\}$ {(x, y) | 3x + y = 1}
- [ ] $\{(x, y) \mid x = 1\}$ {(x, y) | x = 1}
- [ ] $\{(x, y) \mid y = 5x^2\}$ {(x, y) | y = 5x2}
- [ ] $\{(x, y) \mid y = x^3 + 2\}$ {(x, y) | y = x3 + 2}
- [ ] {(1, 2)} {(1, 2)}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(x, y) \mid x^2 + y^2 = 0\}$ {(x, y) | x2 + y2 = 0}

$\{(x, y) \mid 3x + y = 1\}$ {(x, y) | 3x + y = 1}

$\{(x, y) \mid x = 1\}$ {(x, y) | x = 1}

{(1, 2)} {(1, 2)}

1 point

Which of following subsets of R^3 R3 represent an affine subspace of R^3 R3?

- [ ] $\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$ {(x, y, z) | x2 + y2 + z2 = 0}
- [ ] $\{(x, y, z) \mid x^2 + z^2 + y = 1\}$ {(x, y, z) | x2 + z2 + y = 1}
- [ ] $\{(x, y, z) \mid x + y + z = 2\}$ {(x, y, z) | x + y + z = 2}
- [ ] $\{(x, y, z) \mid 2x + y = 3z\}$ {(x, y, z) | 2x + y = 3z}
- [ ] $\{(x, y, z) \mid xy = z\}$ {(x, y, z) | xy = z}
- [ ] $\{(x, y, z) \mid y = x + z\}$ {(x, y, z) | y = x + z}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(x, y, z) \mid x^2 + y^2 + z^2 = 0\}$ {(x, y, z) | x2 + y2 + z2 = 0}

$\{(x, y, z) \mid x + y + z = 2\}$ {(x, y, z) | x + y + z = 2}

$\{(x, y, z) \mid 2x + y = 3z\}$ {(x, y, z) | 2x + y = 3z}

$\{(x, y, z) \mid y = x + z\}$ {(x, y, z) | y = x + z}

1 point

Let U U be a subspace of the vector space R^3 R3 and a basis of U U is given by $\{(0, 1, -3), (1, 0, -1)\}$ {(0, 1, −3), (1, 0, −1)}. Then which of the following subsets of R^3 R3 is an affine subspace of R^3 R3 such that the corresponding vector subspace is U U?

- [ ] $L = \{(x, y, z) \mid x + 3y + z = 3\}$ L = {(x, y, z) | x + 3y + z = 3}
- [ ] $L = \{(x, y, z) \mid x + 3y + 2z = 0\}$ L = {(x, y, z) | x + 3y + 2z = 0}
- [ ] $L = \{(x, y, z) \mid x + 3y + z = 0\}$ L = {(x, y, z) | x + 3y + z = 0}
- [ ] $L = \{(x, y, z) \mid x + 3y + z = 8\}$ L = {(x, y, z) | x + 3y + z = 8}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L = \{(x, y, z) \mid x + 3y + z = 3\}$ L = {(x, y, z) | x + 3y + z = 3}

$L = \{(x, y, z) \mid x + 3y + z = 0\}$ L = {(x, y, z) | x + 3y + z = 0}

$L = \{(x, y, z) \mid x + 3y + z = 8\}$ L = {(x, y, z) | x + 3y + z = 8}

1 point

Which of the following mappings is an affine mapping from R^2 R2 to an affine subspace of R^3 R3?

- [ ] $T(x, y) = (x + 1, y, xy)$ T(x, y) = (x + 1, y, xy)
- [ ] $T(x, y) = (x, y, x + y + 5)$ T(x, y) = (x, y, x + y + 5)
- [ ] $T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$ T(x, y) = (2x − y + 2, 3y + 3, x + y − 1)
- [ ] $T(x, y) = (3x, y^2, x)$ T(x, y) = (3x, y2, x)
- [ ]

$T(x, y) = (x, x + y, 0)$T(x, y) = (x, x + y, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$T(x, y) = (x, y, x + y + 5)$T(x, y) = (x, y, x + y + 5)

$T(x, y) = (2x - y + 2, 3y + 3, x + y - 1)$T(x, y) = (2x − y + 2, 3y + 3, x + y − 1)

$T(x, y) = (x, x + y, 0)$T(x, y) = (x, x + y, 0)

Level 2

1 point

Consider a system of linear equations

$$\begin{aligned} -x + y - z &= 1 \\ x - y + z &= -1 \\ x + z &= 0 \end{aligned}$$
−x + y − z = 1x − y + z = −1x + z = 0

Which of the following is an affine subspace LL of R^3R3 such that if $v \in L$v ∈ L, then vv is a solution of the system of linear equations?

○ $\{(t, -1, t) \mid t \in R\}$\{(t, −1, t) | t ∈ R\}

○ $\{(-t, 1, t) \mid t \in R\}$\{(−t, 1, t) | t ∈ R\}

○ $\{(-t, 1, -t) \mid t \in R\}$\{(−t, 1, −t) | t ∈ R\}

○ $\{(-t, -1, t) \mid t \in R\}$\{(−t, −1, t) | t ∈ R\}

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-t, 1, t) \mid t \in R\}$\{(−t, 1, t) | t ∈ R\}

1 point

Consider the following affine mappings from R^3R3 to an affine subspace of R^3R3

$f_1(x, y, z) = (x + 1, x + 2y - 2, x + y + 2z)$f1(x, y, z) = (x + 1, x + 2y − 2, x + y + 2z)

$f_2(x, y, z) = (x + 5, x + 2y - 3, x + y + 2z + 8)$f2(x, y, z) = (x + 5, x + 2y − 3, x + y + 2z + 8)

$f_3(x, y, z) = (2x + 4, y + z - 1, x + y + 5)$.f3(x, y, z) = (2x + 4, y + z − 1, x + y + 5).

Which of the following option(s) is(are) true?

☐ A linear transformation corresponding to f_1f1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$T(x, y, z) = (x, x + 2y, x + y + 2z)

☐ The linear transformations corresponding to f_1f1 and f_2f2 are not the same.

☐ A linear transformation corresponding to f_3f3 is $T(x, y, z) = (x, y + z, x + y)$T(x, y, z) = (x, y + z, x + y)

☐ A linear transformation corresponding to f_3f3 is $T(x, y, z) = (2x, y + z, x + y)$T(x, y, z) = (2x, y + z, x + y)

No, the answer is incorrect.
Score: 0
Accepted Answers:
A linear transformation corresponding to f_1 f1 is $T(x, y, z) = (x, x + 2y, x + y + 2z)$T(x, y, z) = (x, x + 2y, x + y + 2z)
A linear transformation corresponding to f_3 f3 is $T(x, y, z) = (2x, y + z, x + y)$T(x, y, z) = (2x, y + z, x + y)

Suppose $f : R^2 \rightarrow R^2$ f : R2 → R2 is an affine mapping such that $f(2, 3) = (3, -1)$f(2, 3) = (3, −1), $f(2, 1) = (1, 2)$f(2, 1) = (1, 2) and $f(1, 0) = (0, 1)$f(1, 0) = (0, 1). If $f(-2, 5) = (a, b)$f(−2, 5) = (a, b), then value of $a + b$a + b is

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) -9
1 point

Check Answers
Your score is: 0/8